A Reproducible Benchmark for Zero-Shot Cross-Subject Hand Gesture Recognition Using sEMG Signals

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***Abstract*—This article discusses the use of surface electromyography (sEMG) to identify hand gestures. sEMG is a non–invasive method for recording neuromuscular activity with good temporal resolution, making it suitable for this task. Despite the progress of machine learning and deep learning, the generalizing ability of a model between different people remains challenging due to individual differences. The article presents a reproducible sEMG-based hand gesture recognition experiment under strict conditions where the model is trained on some people and tested on others without any adaptation. Using the NinaPro DB2 [1] open dataset and the LOSO (leave-one-subject-out) estimation method, we compared classical machine learning models and deep learning models working with raw signals and various feature sets. The figures obtained are comparable to data from previous studies, which indicates the complex nature of sEMG gesture recognition when testing is conducted by an unknown user [2]–[4]. Differences between individuals still strongly affect the accuracy of gesture recognition. This means that neither traditional nor advanced teaching methods can yet fully solve the problem of generalizing results for different people. Instead of suggesting a new structure, in this article we present an experiment and an evaluation system that can be fully replicated. They can be a good starting point for future research in the field of reliable and generalizable sEMG-based hand gesture recognition. The proposed criteria and data can help in an objective assessment of future hand gesture recognition methods by different people. The research code is available on github at the following link: https://github.com/Halacky/OMEGA**

***Index Terms*—surface electromyography (sEMG), hand gesture recognition, generalization between subjects, trial-free learning, deep learning, machine learning, LOSO.**

I. Introduction

Surface electromyography (sEMG) is a widely-used non-invasive method for recording neuromuscular activity with high temporal resolution. [5], [6]. Due to its properties, sEMG is used for hand gesture recognition, opening up its applications in many areas of human life [7], [8]. Progress in machine learning and deep learning has significantly improved the accuracy of gesture recognition according to sEMG data [9], [10].

However, most studies evaluate methods using training and testing data obtained from the same person. This often leads to high accuracy, but it overlooks an important issue of sEMG: variability between subjects [11]–[13]. Recent work suggests ways to adapt to specific subjects to solve this problem, but they still require access to end-user data and do not take into account the zero-sample scenario where the model makes predictions without any information about the new subject [14], [15]. Thus, intersubjective assessment remains insufficiently studied, and the lack of standardized and reproducible protocols makes it difficult to directly compare methods and assess their real generalizing ability.

In this article, we fill in the existing gap by offering a reproducible test for recognizing hand gestures based on sEMG signals. The test is designed to evaluate the performance of models on different people under strict conditions. Special attention is paid to the most difficult and useful scenario in practice, when models are trained on a group of people and tested on complete strangers. No prior training, calibration, or adaptation to a specific person is required.

Using the well-known NinaPro DB2 dataset and the LOSO unified protocol, we compare traditional machine learning methods and modern deep learning architectures using different signal representations. The results show that in strict intersubject conditions, deep learning models do not always show better results than well-tuned classical methods. Variability between different people remains the main factor limiting recognition accuracy.

With clear data separation, clear preprocessing, and transparent evaluation criteria, the work provides a realistic assessment of modern sEMG gesture recognition techniques. It creates reproducible conditions that can be used in future generalization studies for different people.

II. Related Work

A. sEMG-Based Hand Gesture Recognition

The first methods of gesture recognition based on sEMG signals relied on manually calculated time and frequency parameters. After that, well-known classifiers were used, for example, SVM or LDA [16], [17]. Although these methods produced good results in gesture classification tasks, their ability to generalize to other subjects was small.

B. Intersubject and Zero-Proof Learning in sEMG

As for interdisciplinary and zero-validation learning in sEMG, several papers have studied in depth the possibility of generalizing results to new objects using protocols that exclude one object [18]–[20]. These studies show a significant decrease in accuracy during the transition to new facilities, in comparison with estimates made at known facilities. This indicates serious problems that arise during intersubject transfer.

It is worth noting that a drop in performance is observed even when using the same model architectures with strict cross-subject evaluation. This indicates that the complexity of the architecture by itself does not guarantee the creation of representations independent of a specific object [21], [22].

C. Engineered Features vs. Learned Representations

Standard time series analysis methods, such as MAV, RMS, WL, ZC, and SSC, are still useful, especially with ensemble classifiers such as random forests [23].

There is evidence that extended feature sets with functions between channels and in the frequency domain can produce results comparable or even better than those of deep learning models [24]. Although deep learning allows for end-to-end tuning, it still strongly depends on the quality of the signal and the decisions made at the preprocessing stage [25].

D. Domain Adaptation and Generalization

These methods, while noticeably improving performance between different areas, require access to target area data [26]. This contradicts the principle of learning from scratch and limits their use when there is no prior information about the target user.

E. Positioning of This Work

Unlike many previous studies, this work focuses on a systematic, fully reproducible evaluation of classical and deep learning methods under a strict zero-trial cross-subject protocol. Using a unified LOSO framework on NinaPro DB2, we:

- enforce strict disjointness between training and test subjects,
- forbid any target-subject data for pre-training or adaptation,
- and provide consistent preprocessing and evaluation metrics.

The purpose of this approach is to separate the actual generalization capability of the model from the distortions caused by specific data sets or differences in procedures.

III. Problem Definition

Let

$$\mathcal{D} = \{(X_i, y_i, s_i)\}_{i=1}^{N} \tag{1}$$

denote the dataset, where $X_i \in \mathbb{R}^{C\times T}$ is a windowed sEMG segment with C channels and T time samples, $y_i \in \{1, \ldots, K\}$ is the corresponding gesture label, and $s_i \in \mathcal{S}$ is the subject identifier.

We partition the subjects into disjoint training and test sets,

$$\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset. \tag{2}$$

The objective is to learn a gesture classification function

$$f : \mathbb{R}^{C\times T} \to \{1, \ldots, K\} \tag{3}$$

using only samples from subjects in $\mathcal{S}_{\text{train}}$, and to evaluate its performance exclusively on samples from unseen subjects in $\mathcal{S}_{\text{test}}$. No data from $\mathcal{S}_{\text{test}}$ is used at any stage of model training, selection, or hyperparameter tuning.

IV. Methods

A. Dataset Description

All experiments were conducted on the publicly available Ninapro DB2 sEMG dataset. The dataset contains multi-channel forearm sEMG recordings collected from 40 healthy subjects performing 49 gestures, 10 repetitions each. Each repetition lasts 5 seconds, with a 3-second rest period between gestures.

The study will exclude signals from the primary activity zone of the flexor and extensor muscles of the superficial fingers, as well as signals from the primary activity zone of the biceps brachii and triceps brachii. Therefore, the data collected from eight sensors evenly distributed around the forearm will be used. This structure was created to ensure the reliability and practical value of the data obtained, due to the fact that it corresponds to the structure of many sEMG wearable bracelets on the market (usually with 8–12 electrodes). This increases the likelihood of using the developed algorithms in real devices.

B. Signal Preprocessing

To reduce typical distortion, the signals are processed in two ways: simple and improved pipelining. The sampling rate of all recordings is 2 kHz.

First, the linear trend is removed to adjust the baseline. Then a 50 Hz notch filter ($Q = 30$) eliminates interference from the power grid, after which a 4th-order Butterworth bandpass filter (20–450 Hz) is applied [27], [28]. The improved method also determines the saturation of the signal using a threshold based on the deviations of the signal from the median of the channel. Values that exceed the threshold are adjusted using linear interpolation of adjacent acceptable values.

Gesture repetitions are segmented using stimulus markers and divided into fixed-length windows. To mitigate the impact of artifacts appearing at the gesture boundaries, we excluded the first and last 10–15% of samples of each repetition. In the process of training deep learning models, z-normalization was applied to each channel based on statistical data obtained exclusively from the training sample.

The enhanced pipeline uses percentile-based normalization (5th–95th percentiles) [29] whose values are clipped to $\pm 3\sigma$ to limit the impact of outliers. For classical machine learning methods, normalization is applied after feature extraction using the same scheme.

C. Feature Representations

The paper considers four ways of representing features.

1) Source Signal Windows: Windows of the original signal: Each window has the form of a tensor $X \in \mathbb{R}^{C \times T}$, where $C = 8$ channels, and T = from 500 to 600 time samples.

2) Time-Domain Features: Time-domain features: For each channel, five standard time domain indicators are calculated, such as:

- mean absolute value (MAV),
- root mean square (RMS),
- waveform length (WL),
- zero crossings (ZC),
- slope sign changes (SSC).

The resulting feature vector concatenates all channels.

3) Sequential Temporal Features: Time signs in the form of a sequence: The window is divided into overlapping fragments of 50 ms in increments of 25 ms. For each fragment and channel, temporary features are extracted, forming a sequence of feature vectors of size $(F \times 40)$, where F is the number of fragments in the window.

4) Advanced Handcrafted Features: Extended features generated manually: A complete set of features is obtained by combining features of the time, frequency, and time-frequency domains.

For conventional machine learning models, PCA or SelectKBest are additionally used to reduce dimensionality.

V. Models and Training Protocol

A. Deep Learning Models

Convolutional models: Standard one-dimensional convolutional neural networks (CNNs) are used. To assess their ability to adaptively recalibrate informative features, options involving mechanisms of channel and spatial attention are being considered. Also, CNN architectures with parallel convolutional cores allow capturing temporal patterns with different resolutions [30], [31].

Recurrent models: Bidirectional LSTM and GRU networks are used to model time dependencies in sEMG windows [32]. Attention-based extensions are included to highlight important time stages in the formation of general ideas [33].

Temporal Convolutional Networks: TCN models with dilated convolutions are evaluated for their ability to model long-term temporal dependencies. A variant of TCN with a dilated attention mechanism, augmenting hierarchical temporal abstraction with global self-attention, is also tested.

Hybrid models: For structured and multidimensional feature representations created by hand, hybrid architectures combining convolutional and recurrent layers, as well as fully connected networks, are used [34], [35].

B. Traditional Machine Learning Models

The support vector machine (SVM) is used with linear and radial basis function (RBF) kernels. For each training block, hyperparameters are selected based on the validation data.

Random forest classifiers are used because of their resistance to redundancy and multidimensional input spaces. The models are trained using 300 trees and a balanced weight of classes [36].

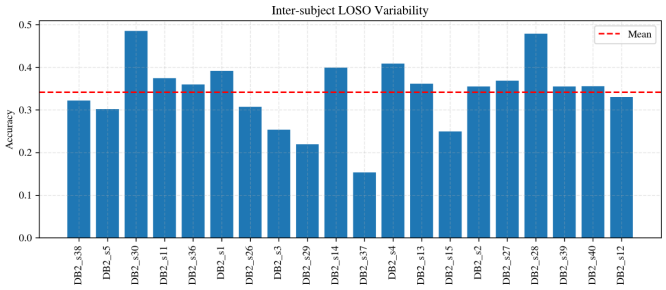

Fig. 1. Accuracy per subject.

C. Training and Evaluation Protocol

All deep learning models are trained with Adam, the initial learning rate is 10^{-3} and the weight decay is 10^{-4}. The batch size ranges from 256 to 4096, depending on the hardware. The gradient norms are clipped to 5.0. Cross-entropy is used, and class weighting is taken into account if necessary. The training lasts for 50 epochs, with early stopping (7 epochs of patience) based on the validation loss. Dropout is applied to the selected layers with a probability of $p = 0.3$, batch normalization is used for learning stability.

For reproducibility, random seeds are fixed throughout the pipeline. Python, NumPy, PyTorch, and CUDA are used. Whenever possible, deterministic operations are used.

Evaluation follows a strict LOSO protocol. For each subject, models are trained exclusively on data from all other subjects and evaluated on the held-out subject. Validation data is drawn only from the training subjects. Final performance is reported as mean and standard deviation of classification accuracy and macro-F1 across all test subjects.

VI. Experimental Results

A. Untrained Baseline Performance

Initially, a simple convolutional neural network (CNN) was developed and trained on the initial EMG data using the LOSO protocol. The average accuracy was 35.9% $\pm$ 8.5%, and macro-F1 was 33.3% $\pm$ 9.1% for unseen participants.

The results showed a spread between the participants (accuracy from 21% to 52%). An analysis of the results for each gesture Fig. 1) indicated a different complexity of classification, with F1 varying from 0.26 to 0.43. It was noted that some gestures were classified better than others.

Some gestures are recognized relatively well (e.g., gestures 15–17), while others remain difficult. This suggests that both intersubject and intergesture variability play a role in the observed performance.

B. Architecture Comparison

Table I summarizes the performance of all studied architectures under the LOSO protocol on a subset of 20 randomly selected subjects and gestures (no biased selection). Results

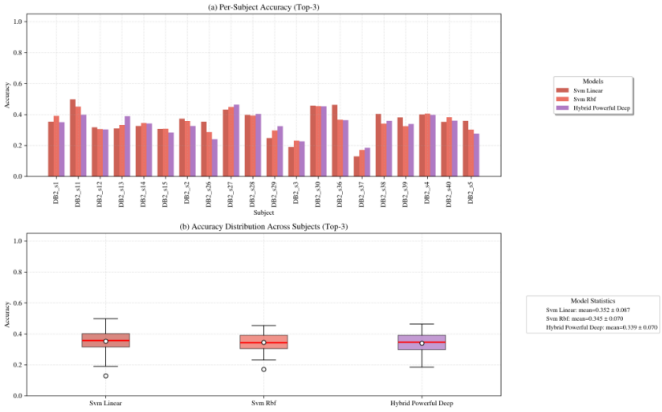

Fig. 2. Subject-wise accuracy under the LOSO protocol.

are reported for models trained on raw sEMG and on time-domain (TD) features.

The analysis showed that among the models working with raw data, a simple convolutional network achieves the best average accuracy (35.9%). Recurrent-type models and models based on attention mechanisms do not show a steady improvement in results.

Models trained using TD functions are characterized by a smaller spread of results between different tasks, although their maximum performance is somewhat reduced. Traditional machine learning methods, in particular linear SVM, demonstrate performance comparable to the best deep learning models, while there is no statistically significant difference in average accuracy.

C. Leave-One-Subject Analysis

1) Intersubject Variability: A subject-wise analysis (Fig. 2) reveals substantial intersubject variability. Different models show different accuracy, which does not depend on the architecture. Some models are always less accurate, while others are more accurate. At the same time, the relative rating of the models does not change depending on the region. This suggests that the choice of architecture improves the situation, but not always.

2) Statistical Significance: Paired t-tests among the best-performing deep learning models show that a CNN-GRU model with attention yields statistically significant improvements over several weaker baselines ($p < 0.05$). However, differences between the remaining top-performing models are statistically insignificant, and the performance gap between deep learning and well-tuned classical models (e.g., linear SVM) is also small.

D. Summary of Results

Overall, while some architectural changes (e.g., combining convolution and recurrent layers with attention) provide moderate gains, cross-subject performance remains primarily constrained by intersubject variability rather than model capacity. Classical machine learning models, when equipped with appropriate features and normalization, remain highly competitive in this strict zero-trial setting.

VII. Discussion

The presented results indicate the difficulties of recognizing gestures by EMG in different people when learning without examples. None of the tested neural networks has shown a clear advantage over conventional machine learning methods. Most likely, the differences between people are more important than choosing a specific model.

These results show that surface EMG has its limitations as a method. Even if the data is collected in the same way, small changes in the location of the electrodes and personal characteristics of people strongly affect the results.

In this study, we used a complex but realistic approach: learning without examples. This means that we did not customize the system for each person. Although the recognition accuracy is not very high, we can honestly assess how well the system works for different people.

VIII. Limitations and Future Work

Despite the systematic assessment, it is worth noting some limitations. Firstly, the experiments were conducted only on NinaPro DB2. Secondly, the study deliberately did not use adaptation and personalization methods in order to focus on the scenario without preliminary tests.

Also, due to hardware and computing limitations, the assessment was limited to a set of gestures and participants. Gestures and participants were selected randomly, without trying to select favorable cases, which could affect performance indicators.

Future studies will expand the benchmark to include multiple datasets with different data collection protocols and electrode locations.

IX. Conclusion

This study offers a detailed and reproducible assessment of sEMG-based hand gesture recognition techniques when there is no data available on specific individuals. The results show that recognizing gestures between different people is a difficult task, where the main limitations are the variability of the signal and the characteristics of each person, rather than the model itself.

This work introduces a reproducible benchmark for cross-subject sEMG gesture recognition, providing the first comprehensive comparison of 10+ architectures under strict LOSO protocol and demonstrating that intersubject variability, not model complexity, is the primary performance bottleneck. By evaluating models under harsh generalization conditions, this work aims to provide a more realistic assessment of progress in sEMG-based gesture recognition and can serve as a basis for future research on reliable generalization between different people.

TABLE I
SUMMARY OF MODEL PERFORMANCE (LOSO, $N = 20$ SUBJECTS). VALUES ARE MEAN ± STD ACROSS SUBJECTS.

Model Category	Architecture	Features	Accuracy (%)	F1-Score
Raw-Signal DL	CNN/LSTM BiGRU	Raw EMG	35.7 ± 7.8	33.5 ± 8.0
	Multiscale CNN	Raw EMG	35.5 ± 8.1	33.4 ± 8.8
	Simple CNN	Raw EMG	**35.9 ± 8.5**	33.3 ± 9.1
	CNN-GRU-Attn	Raw EMG	34.8 ± 8.8	33.1 ± 9.1
	Attention CNN	Raw EMG	34.2 ± 8.0	31.1 ± 8.7
	BiLSTM-Attn	Raw EMG	34.0 ± 7.8	31.9 ± 8.1
	TCN	Raw EMG	26.7 ± 7.4	23.1 ± 6.9
	TCN-Attn	Raw EMG	25.5 ± 13.5	20.8 ± 16.0
	BiLSTM	Raw EMG	21.4 ± 10.4	16.0 ± 12.6
Feature-Based DL	CNN/LSTM BiGRU	TD Features	32.7 ± 7.2	30.7 ± 8.0
	TCN-Attn	TD Features	31.9 ± 6.7	30.2 ± 7.0
	BiLSTM-Attn	TD Features	30.8 ± 6.4	29.4 ± 7.0
	BiLSTM	TD Features	30.7 ± 6.3	29.2 ± 6.9
	CNN-LSTM	TD Features	29.8 ± 5.5	27.9 ± 5.7
	Simple CNN	TD Features	27.9 ± 5.3	25.6 ± 5.9
Classical ML	SVM-Linear	TD Features	**35.2 ± 8.9**	**32.5 ± 9.4**
	SVM-RBF	TD Features	34.5 ± 7.2	32.6 ± 7.7
	Random Forest	TD Features	32.0 ± 6.5	30.3 ± 7.0
Hybrid DL	Hybrid Powerful Deep	TD Features	33.9 ± 7.2	31.7 ± 7.0
	MLP Powerful	TD Features	33.8 ± 7.4	31.6 ± 7.3

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